SMART INDIA HACKATHON 2026

OceanLens 3D

Problem Statement ID – 26067

Problem Statement Title – Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations

Theme – Disaster Management

PS Category – Software

Team ID – <your team ID from the portal>

Team Name – ByteForge



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OCEANLENS 3D

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Proposed Solution (Describe your Idea/Solution/Prototype)

WHAT IT IS

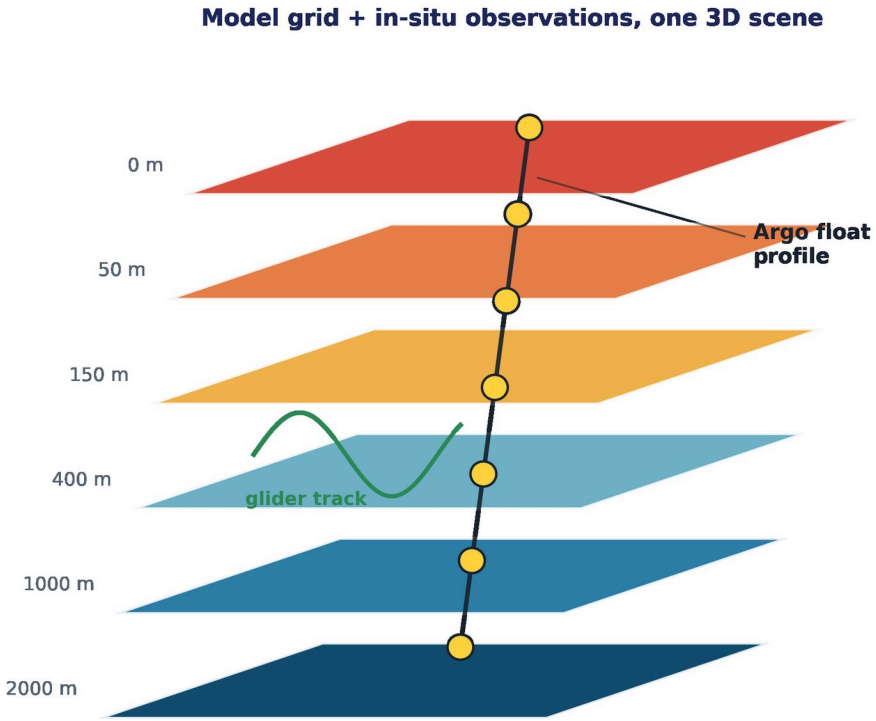
- One self-contained HTML file (~3 MB) that opens the Arabian Sea and Bay of Bengal (6–22°N, 61–92°E) as a navigable 3D ocean volume
- No install, no login, no server — runs offline from a pen drive or an email attachment
- Real case study built in: Cyclone Amphan, May 2020 — HYCOM model analyses, 206 Argo profiles (with BGC chlorophyll and oxygen) and the IBTrACS track

HOW IT ADDRESSES THE PROBLEM

- Model fields (temperature, salinity, currents) and in-situ Argo, BGC, glider and CTD profiles are drawn in the SAME 3D scene instead of separate portals
- Click any float → its observed profile is drawn over the co-located model column with bias and RMSE, so model–observation agreement is visible in one glance
- Depth slices, vertical sections, 3D isosurfaces, D20 / D26 surfaces, a cyclone heat potential map, current particle streaks, ETOPO1 bathymetry and coastline

INNOVATION AND UNIQUENESS

- Existing tools (Ocean Navigator, MyOcean Pro, Argovis) need servers, accounts or live bandwidth; OceanLens 3D needs a browser; drag in your own CSV or NetCDF files
- Density (EOS-80), sound speed (Mackenzie 1981) and cyclone heat potential computed live in the browser — not pre-rendered pictures



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TECHNICAL APPROACH

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TECHNOLOGIES USED

FRONT END
HTML5 · vanilla JavaScript
Three.js r128 (WebGL)

DATA LAYER
Python · NumPy
NetCDF / CF · OPeNDAP

OCEANOGRAPHY
UNESCO/EOS-80 density
Mackenzie 1981 sound speed

VALIDATION
Argo QC flags + density test
bias · RMSE at each profile's time

METHODOLOGY — DATA FLOW

DATA SOURCES
HYCOM GOFS 3.1 analysis
Argo + BGC-Argo (ERDDAP)
ETOPO1 relief
IBTrACS best tracks
next: INCOIS LAS · Copernicus

→

INGESTION (Python)
OPeNDAP · ERDDAP · CF NetCDF
0.5° regrid, 11 depths
Argo QC, pressure → depth
16-bit quantize

→

PAYLOAD
16-bit arrays, base64
+ JSON header
~2.3 MB for 8 analyses

→

BROWSER CLIENT
one .html file
CSV + NetCDF import
no install, offline

RENDERING METHOD

- Volume drawn as an opaque sea surface over faint depth layers (alpha blending, back to front) — stable on integrated GPUs where ray-marching fails
- Marching-cubes isosurfaces and D20 / D26 surfaces · log or linear colour scales · particle advection for currents · land masking against ETOPO1
- Status: working prototype on real Cyclone Amphan data — volume, slices, sections, isosurfaces, float profiles, file import, plugins, REST / WMS / WCS server

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FEASIBILITY AND VIABILITY

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WHY IT IS FEASIBLE

✓ Prototype exists
Full 3D scene, float comparison and demo mode already run on real Cyclone Amphan data in one ~3 MB file, built by a working Python pipeline and checked by 159 automated tests.

✓ Data is free and open
HYCOM, Argo GDAC, ETOPO1 and IBTrACS are public: no account needed, free to use with attribution; Copernicus Marine needs only a free registration.

✓ Runs on any laptop
Browser + WebGL only. No GPU cluster, no cloud bill, no IT deployment for a district office to use it.

✓ Built-in quality checks
Argo QC flags, a density-inversion test and bias / RMSE against the model run inside the app, so bad data is flagged, not hidden.

RISKS → MITIGATION

Payload size vs. detail
Regional subsetting + vertical regridding + 16-bit quantization keeps 8 analyses of model fields plus 206 Argo profiles to about 2.3 MB.

Weak GPUs on office laptops
Layered transparency instead of ray-marching: 11 textured depth layers, cheap on integrated GPUs; Three.js version pinned.

Sparse / gappy float coverage
Gaps are shown as gaps, never silently interpolated, so users are not misled about data density.

Long-term maintenance
One pinned library and a standard-library build, no npm chain; a plugin API adds sensors and variables without touching the core.

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IMPACT AND BENEFITS

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0
installs, logins or servers needed

~3 MB
app + real data, opens offline

5 → 1
data sources in a single 3D view

IMPACT ON THE TARGET AUDIENCE

INCOIS scientists
Model–observation comparison in seconds instead of hours; an optional server publishes the same data over REST, OGC WMS/WCS and OPeNDAP.

Cyclone & disaster cells
D26 depth and cyclone heat potential mapped along the storm track and exported for a briefing before landfall; the Amphan 2020 case is built in.

Coastal & fisheries offices
Usable on low-bandwidth or offline machines where existing web portals simply fail to load.

Students & educators
Guided tour mode turns a research tool into an ocean-literacy resource for any classroom.

BENEFITS OF THE SOLUTION

Social
Cyclone-relevant ocean heat information reaches district and coastal users, not only well-connected research labs.

Economic
No licence fees, hosting or hardware cost; one file replaces a deployed visualization stack for most inspection tasks.

Environmental
Faster reading of thermocline, eddy and monsoon-reversal structure supports marine and fisheries decision-making.

Scientific
Validation is built in — GODAE Class 4 metrics and density-inversion checks show how far each view can be trusted.

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RESEARCH AND REFERENCES

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DATA SOURCES

- HYCOM GOFS 3.1 global ocean analysis — tds.hycom.org
- Argo and BGC-Argo Global Data Assembly Centre, via Ifremer ERDDAP — erddap.ifremer.fr
- ETOPO1 global relief model (NOAA NCEI) — ncei.noaa.gov
- IBTrACS v04 cyclone best tracks (Knapp et al., 2010) — ncei.noaa.gov
- Next: INCOIS Live Access Server (las.incois.gov.in) and Copernicus Marine (marine.copernicus.eu)

SCIENCE AND METHODS

- UNESCO / EOS-80 international equation of state of seawater (1980)
- Mackenzie, K. V. (1981) — nine-term equation for sound speed in the oceans, JASA 70(3)
- GODAE OceanView Class 4 forecast verification framework
- Goni et al. — tropical cyclone heat potential and the depth of the 26 °C isotherm (D26)
- Leipper & Volgenau (1972) — hurricane heat potential
- Argo Quality Control Manual — density inversion test
- Three.js (threejs.org) · OGC WMS 1.3.0 / WCS 2.0.1 · CF Conventions · OPeNDAP DAP2

EXISTING TOOLS REVIEWED — AND WHAT OCEANLENS 3D DOES DIFFERENTLY

DFO Ocean Navigator (needs a server deployment) · Copernicus MyOcean Pro (account + connectivity) · Argovis (float profiles, not integrated 3D model volumes) · OceanOPS (network status, not ocean fields)

None of them opens from a pen drive on an offline coastal-office laptop and shows a navigable 3D ocean. That portability is our contribution.